

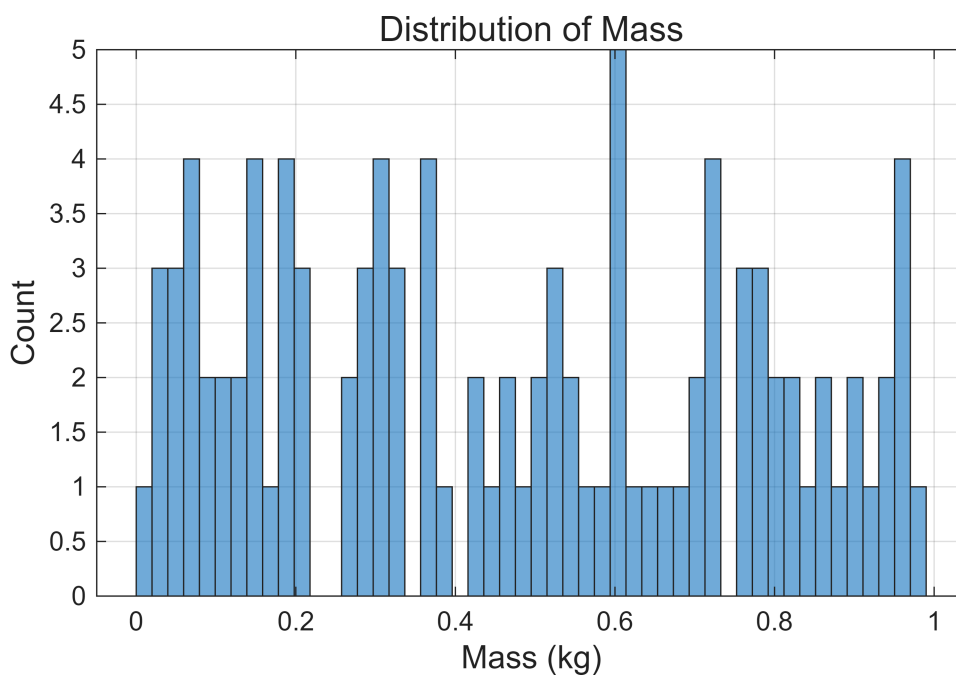
Homework 2: Comparison of Triplet and Stacking Weighing Methods in Model Estimation

4 April 2026 by Ji,Haozheng

Using the Triplet Weighing Method

(a) Randomly assigns masses m_{true} to the objects in the range of 0-1 kg

```
clc; clear; close all;
rng(42);
n_m=100;
m_true = rand(n_m,1);
figure;
histogram(m_true, 50);
xlabel('Mass (kg)', 'FontSize', 12);
ylabel('Count', 'FontSize', 12);
title('Distribution of Mass', 'FontSize', 13);
grid on;
box on;
exportgraphics(gcf, 'mass_distribution.png', 'Resolution', 300);
```



(b) Build the appropriate kernel G

step 1. weight the first

step 2. weight the first and second

step n. weight the rest in triplets

```
G_tri=eye(n_m,n_m);
G_tri(2,1)=1;
```

```

for ii = 3:1:n_m
    G_tri(ii,ii-2:ii-1)=1;
end

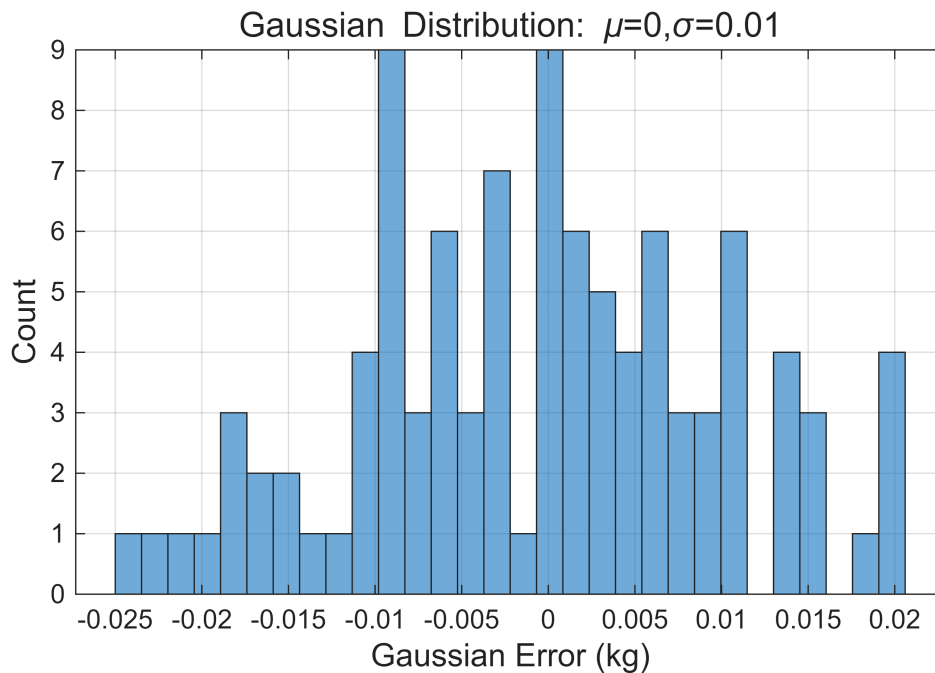
```

(c) Creates synthetic observed data $d_{obs} = Gm + n$, where n is a vector of Gaussian random number with zero mean and $\sigma_d = 0.01kg$

```

mu=0;
sigma=0.01;
n=mu+sigma*randn(n_m,1);
d_tri=G_tri*m_true+n;
figure;
histogram(n,30);
xlabel('Gaussian Error (kg)', 'FontSize', 12);
ylabel('Count', 'FontSize', 12);
title('Gaussian Distribution: \mu=0,\sigma=0.01','FontSize',13)
grid on;
box on;
exportgraphics(gcf, 'error_distribution.png', 'Resolution', 300);

```



(d) Solves the inverse problem by simple least square

$$Gm = d_{obs}, m_{inv} = (G^T G)^{-1} G^T d_{obs}$$

```

G_gtri=(G_tri'*G_tri)\G_tri';
m_tri=G_gtri*d_tri;

```

(e) Estimates the variance of each of the estimated model parameters m_{est}

$$m_{est} = G_g d, \text{ cov}[m] = G_g \text{cov}[d] G_g^T, \text{ and we have } \text{cov}[d] = 0.01^2.$$

The diagonal element are the corresponding variance of model parameters.

```
covar_tri=sigma^2*(G_gtri*G_gtri');
var_tri=diag(covar_tri);
sigma_tri=sqrt(var_tri);
```

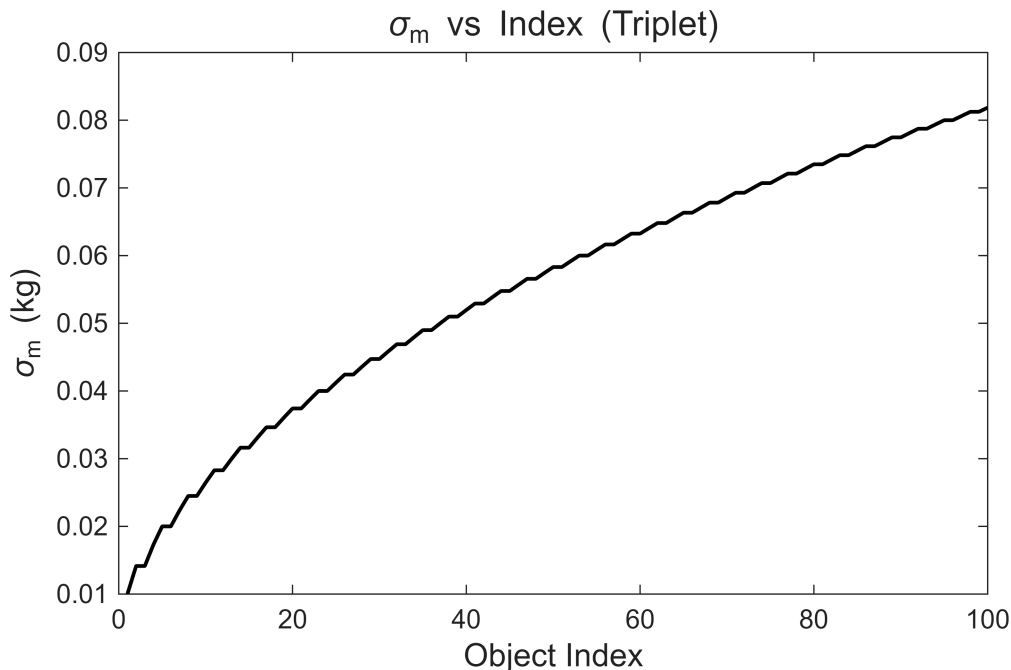
(f) Counts up the number of estimated model parameters that are within $\pm 2\sigma_m$

```
res_tri=m_tri-m_true;
counts_tri=sum(abs(res_tri)-2*sigma_tri<=0);
fprintf('Triplet Weighing Method: Number of parameters within  $\pm 2\sigma_m$  of true
value: %d / %d\n', ...
        counts_tri, n_m);
```

Triplet Weighing Method: Number of parameters within $\pm 2\sigma_m$ of true value: 98 / 100

(g) Make a plot of σ_m as a function of the index of the model parameters

```
figure;
idx=1:n_m;
plot(idx,sigma_tri,'k',LineWidth=1.4);
xlabel('Object Index','FontSize', 12);
ylabel('\sigma_m (kg)','FontSize', 12);
ax.YAxis.Exponent = -3;
title('\sigma_m vs Index (Triplet)','FontSize', 13);
exportgraphics(gcf, 'Deviation_index_Triplet.png', 'Resolution', 300);
```

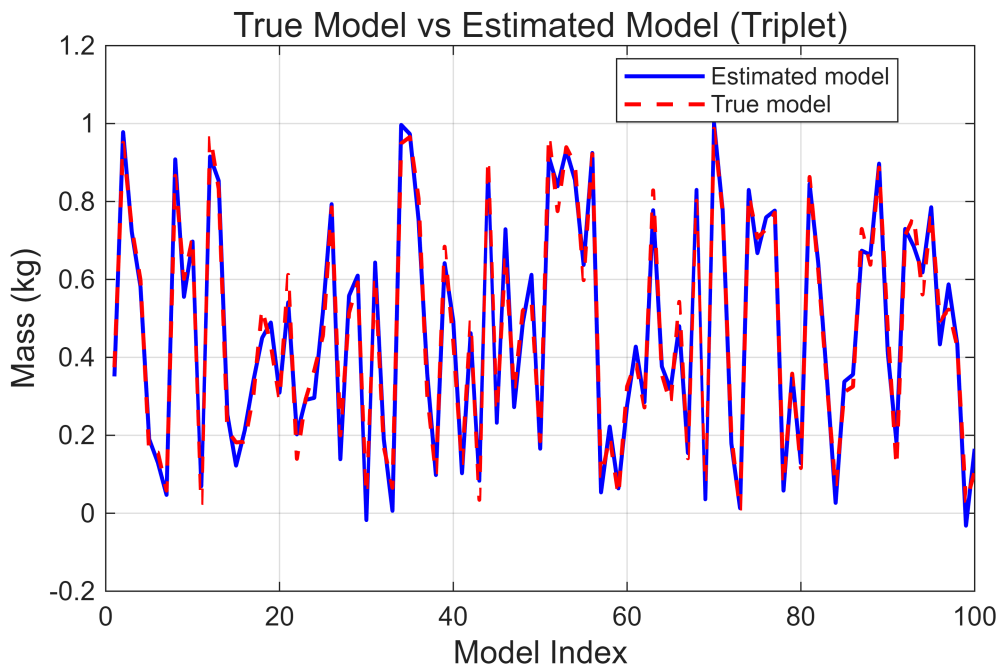


```
figure;
hold on;
plot(idx, m_tri, 'b-', 'LineWidth', 1.5);
plot(idx, m_true, 'r--', 'LineWidth', 1.5);
xlabel('Model Index','FontSize', 12);
ylabel('Mass (kg)','FontSize', 12);
title('True Model vs Estimated Model (Triplet)','FontSize', 13);
legend('Estimated model','True model','Location','best');
```

```

grid on;
box on;
exportgraphics(gcf, 'model_comparison_Triplet.png', 'Resolution', 300);

```



Using the Stacking Weighing Method

step 1. weight the first

step 2. weight the first and second

step n. step n. weigh all objects up to n

```

G_stack=tril(ones(n_m));
d_stack=G_stack*m_true+n;
G_gstack=(G_stack'*G_stack)\G_stack';
m_stack=G_gstack*d_stack;
covar_stack=sigma^2*(G_gstack*G_gstack');
var_stack=diag(covar_stack);
sigma_stack=sqrt(var_stack);
res_stack=m_stack-m_true;
counts_stack=sum(abs(res_stack)-2*sigma_stack<=0);
fprintf('Stacking Weighing Method: Number of parameters within ±2σ_m of true
value: %d / %d\n', ...
        counts_stack, n_m);

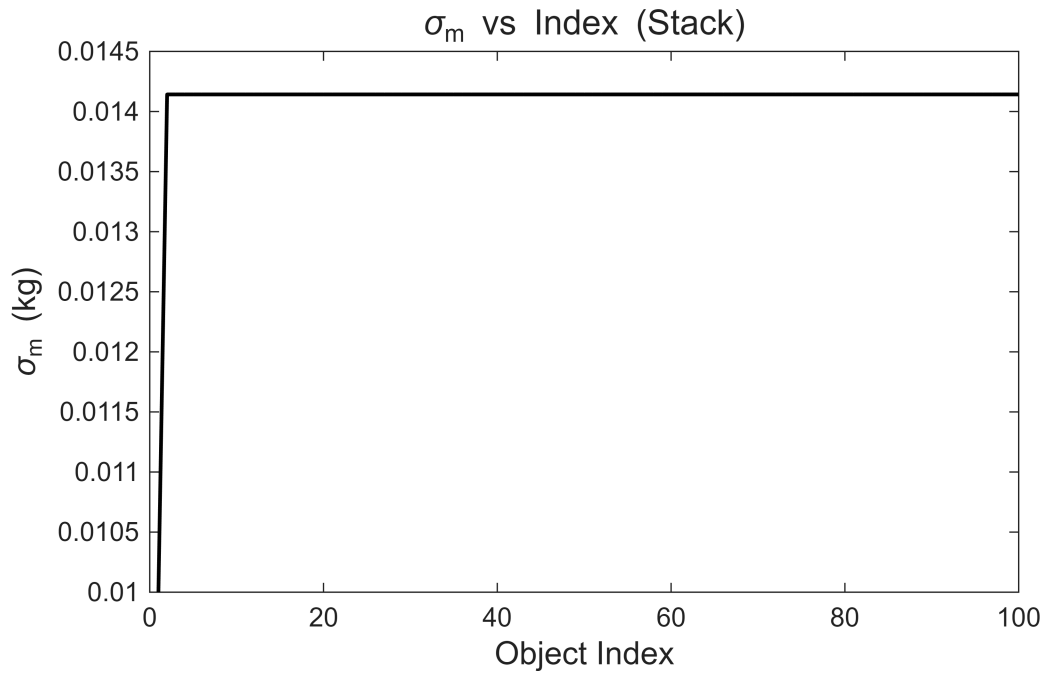
```

Stacking Weighing Method: Number of parameters within $\pm 2\sigma_m$ of true value: 96 / 100

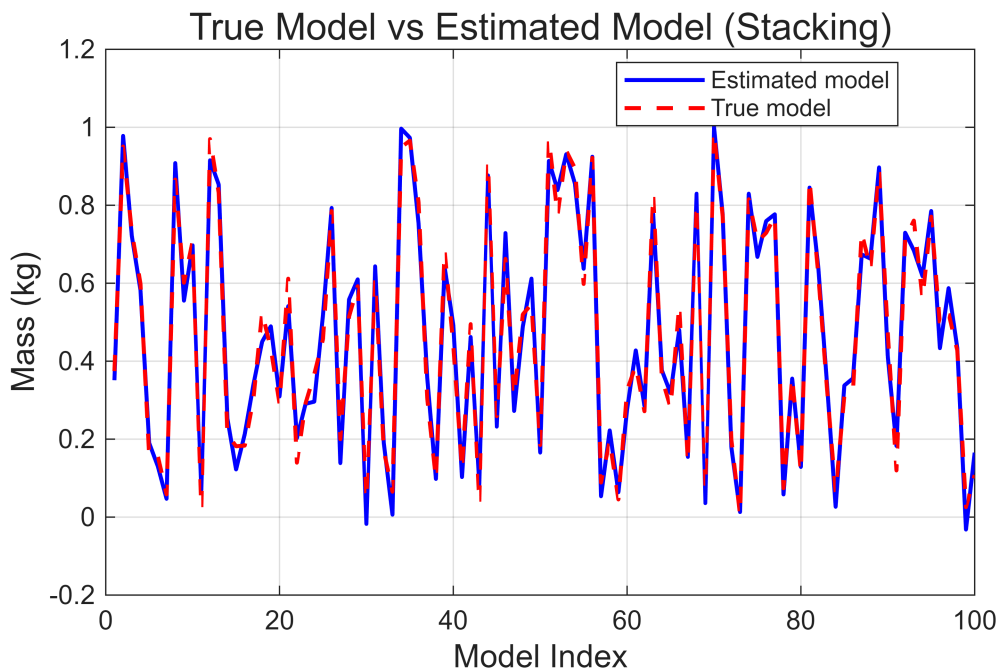
```

figure;
idx=1:n_m;
plot(idx,sigma_stack,'k',LineWidth=1.4);
xlabel('Object Index','FontSize', 12);
ylabel('\sigma_m (kg)','FontSize', 12)
title('\sigma_m vs Index (Stack)','FontSize', 13)
exportgraphics(gcf, 'Deviation_index_Stacking.png', 'Resolution', 300);

```



```
figure;
hold on;
plot(idx, m_tri, 'b-', 'LineWidth', 1.5);
plot(idx, m_true, 'r--', 'LineWidth', 1.5);
xlabel('Model Index','FontSize', 12);
ylabel('Mass (kg)','FontSize', 12);
title('True Model vs Estimated Model (Stacking)','FontSize', 14);
legend('Estimated model','True model','Location','best');
grid on;
box on;
exportgraphics(gcf, 'model_comparison_Stacking.png', 'Resolution', 300);
```



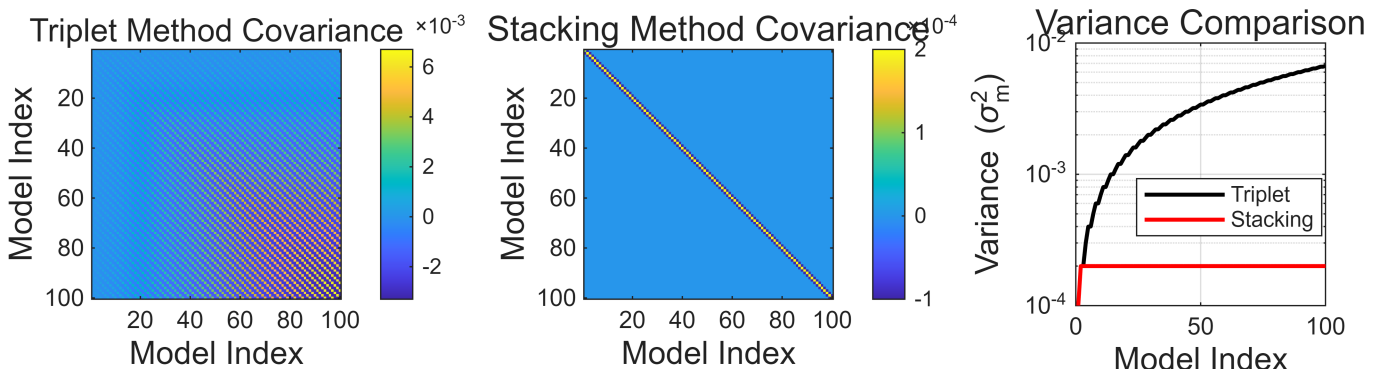
Comparison of Triplet and Stacking Weighing Methods

```
fprintf('Average  $\sigma_m$  - Triplet: %.6f kg\nAverage  $\sigma_m$  - Stacking: %.6f kg\nRMSE - Triplet: %.6f kg\nRMSE - Stacking: %.6f kg\n',
mean(sqrt(var_tri)),mean(sqrt(var_stack)),sqrt(mean(res_tri.^2)),sqrt(mean(res_stack.^2)));
```

```
Average  $\sigma_m$  - Triplet: 0.055201 kg
Average  $\sigma_m$  - Stacking: 0.014101 kg
RMSE - Triplet: 0.040437 kg
RMSE - Stacking: 0.014222 kg
```

Compare the covariance and model resolution

```
figure('Position',[100,100,1400,400]);
t = tiledlayout(1,3,'TileSpacing','Compact','Padding','Compact');
nexttile
imagesc(covar_tri);
colorbar;
axis image;
xlabel('Model Index','FontSize',12);
ylabel('Model Index','FontSize',12);
title('Triplet Method Covariance','FontSize',12);
nexttile
imagesc(covar_stack);
colorbar;
axis image;
xlabel('Model Index','FontSize',12);
ylabel('Model Index','FontSize',12);
title('Stacking Method Covariance','FontSize',13);
nexttile
hold on
plot(1:n_m, var_tri, 'k-', 'LineWidth', 1.5);
plot(1:n_m, var_stack, 'r-', 'LineWidth', 1.5);
xlabel('Model Index','FontSize',12);
ylabel('Variance ( $\sigma_m^2$ )','FontSize',12);
title('Variance Comparison','FontSize',13);
legend('Triplet','Stacking','Location','best');
set(gca,'YScale','log');
grid on;
box on;
exportgraphics(gcf, 'covariance_comparison.png', 'Resolution', 300);
```



Analysis

When using the Stacking Method, each measurement includes more objects, but the measurement deviation is fixed. The more objects included in a measurement, the smaller the portion of error shared by each object, approximately $\frac{\epsilon}{N}$ which decreases toward zero as $N \rightarrow \infty$. In contrast, in the Triplet Method, each measurement always involves three objects, and the error is shared equally among them (approximately $\frac{\epsilon}{3}$), which does not decrease as the model grows. As a result, the Triplet Method accumulates more uncertainty for later objects, while the Stacking Method produces more stable and smaller variance for all estimated masses.

Resolution

- Triplet Method: Covariance matrices (also can be written as $G^T G$) are **scattered** showing many model parameters contribute to one estimated value, resulting in **poor resolution**.
- Stacking Method: Covariance matrices are **approximately diagonal** indicating each estimated parameter is mainly determined by its corresponding true value, giving **high resolution**.

Root Mean Square Error (RMSE)

- Trend: For the Triplet Method, RMSE **increases** with the number of model parameters (N), indicating accuracy deteriorates for larger models.
- Size: Triplet RMSE is generally **larger** than Stacking RMSE, while Stacking RMSE remains **small and stable** across all model sizes.

Standard Deviation (σ_m)

- Trend: For the Triplet Method, σ_m **increases** with N , reflecting that uncertainty accumulates for later parameters. For the Stacking Method, σ_m remains **stable**, showing consistent uncertainty across all parameters.
- Size: Triplet σ_m is generally **larger** than Stacking σ_m , which is **smaller** and well-controlled.

N	RMSE_Triplet	RMSE_Stack	σ_m Triplet	σ_m Stack
10	0.014499	0.011399	0.019341	0.013728
100	0.040437	0.014222	0.055201	0.014101
1000	0.097625	0.014445	0.172386	0.014138
10000	0.559091	0.014009	0.544412	0.014142

